

SECURE SOMEONE ELSE'S COMPUTER: A HANDS-ON CLOUD SECURITY DEMO

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AGENDA

- 1 Why the cloud is everywhere
- 2 **Build** – your first server, *live*
- 3 **Break** – how cloud breaches happen
- 4 **Defend** – seeing the whole battlefield, *live*
- 5 **Design and operationalize** – make security the default
- 6 Q&A



A dramatic sky with a bright sun, blue rays, and large white clouds. The sun is positioned in the upper center, casting a strong glow and creating a lens flare effect. The sky is a deep blue, and the clouds are large, white, and fluffy, with some darker blue shadows on their sides. The overall mood is bright and expansive.

WHO USED THE CLOUD TODAY?

WHY THE CLOUD IS EVERYWHERE



Enterprise top players require infrastructure – compute power, networking, storage capabilities – to ***provide services and achieve business value***.

Software provisioning models shifted ***from owning to renting assets***:

~1995
On-Premises

Companies own and run their servers

~2006
The Shift

Big providers start renting computers by the hour

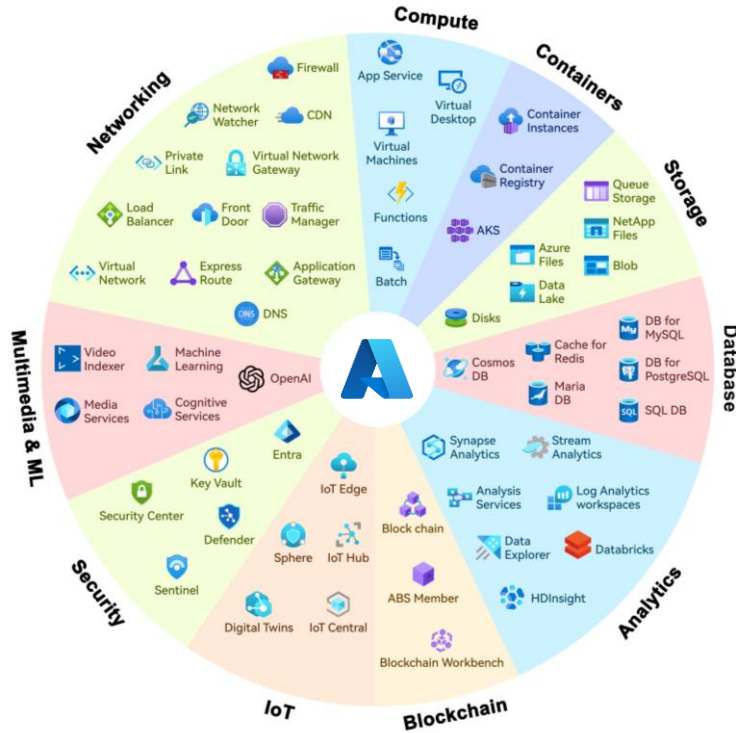
Today
Cloud-Native

Computing as an on-demand, pay-per-use service

aws



WHY THE CLOUD IS EVERYWHERE



	Responsibility	SaaS	PaaS	IaaS	On-prem
Responsibility always retained by the customer	Information and data	■	■	■	■
	Devices (Mobile and PCs)	■	■	■	■
	Accounts and identities	■	■	■	■
Responsibility varies by type	Identity and directory infrastructure	■	■	■	■
	Applications	■	■	■	■
	Network controls	■	■	■	■
	Operating system	■	■	■	■
	Physical hosts	■	■	■	■
Responsibility transfers to cloud provider	Physical network	■	■	■	■
	Physical datacenter	■	■	■	■

■ Microsoft
 ■ Customer
 ■ Shared

Security of the Cloud is mostly up to the CSP
Security in the Cloud is mostly up to the user



BUILD – YOUR FIRST SERVER, LIVE

EVERY CLICK IS A SILENT SECURITY DECISION

Security decisions when creating a VM:

- ***Location*** – Region
- ***Specification*** – Size and running OS
- ***Network exposure*** – who can reach it
- ***Identity and access*** – who can log in



AZURE DEMO



BREAK – HOW CLOUD BREACHES HAPPEN

VMs

- Exposed management ports (SSH, RDP)
- Unpatched OS and software
- Stolen SSH keys

...



Storage

- Public buckets exposing sensitive data
- Leaked access keys

...



Containers

- Vulnerable container images
- Exposed APIs
- Privilege escalation

...



Design and operational mistakes rather than clever hacking:

- ***Misconfiguration of services***
- ***Access perimeters not following least privilege principle***
- ***Unpatched OS and software vulnerabilities***



DEFEND – SEEING THE WHOLE BATTLEFIELD, LIVE

Three major real-world problems:

- **Visibility** – knowing what exists
- **Prioritization** – knowing what matters
- **Multi-cloud, real-time coverage** – handling complex environments relying on several CSPs

Real-world, production-ready solutions include **Cloud Native Application Protection Platforms (CNAPPs)**, namely ***unified platforms to protect enterprise workloads, configurations, identities and data across one or more clouds.***



WIZ DEMO



DESIGN AND OPERATIONALIZE – MAKE SECURITY THE DEFAULT

Three key security enablers:

- ***Security-by-design and shift left*** – security is key epochs before deployment
- ***Prioritization*** – once deployed, focus on what matters most
- ***Business processes*** – in the cloud and AI native world, technology is no more the real bottleneck, human decisions are

Effective cloud security goes beyond mitigating complex attack vectors....

It focuses on ***securely designing and scaling infrastructure to support business logic.***

The core objective is simple: ***identify what the business cannot afford to lose and build environments where the secure path is the default path.***



